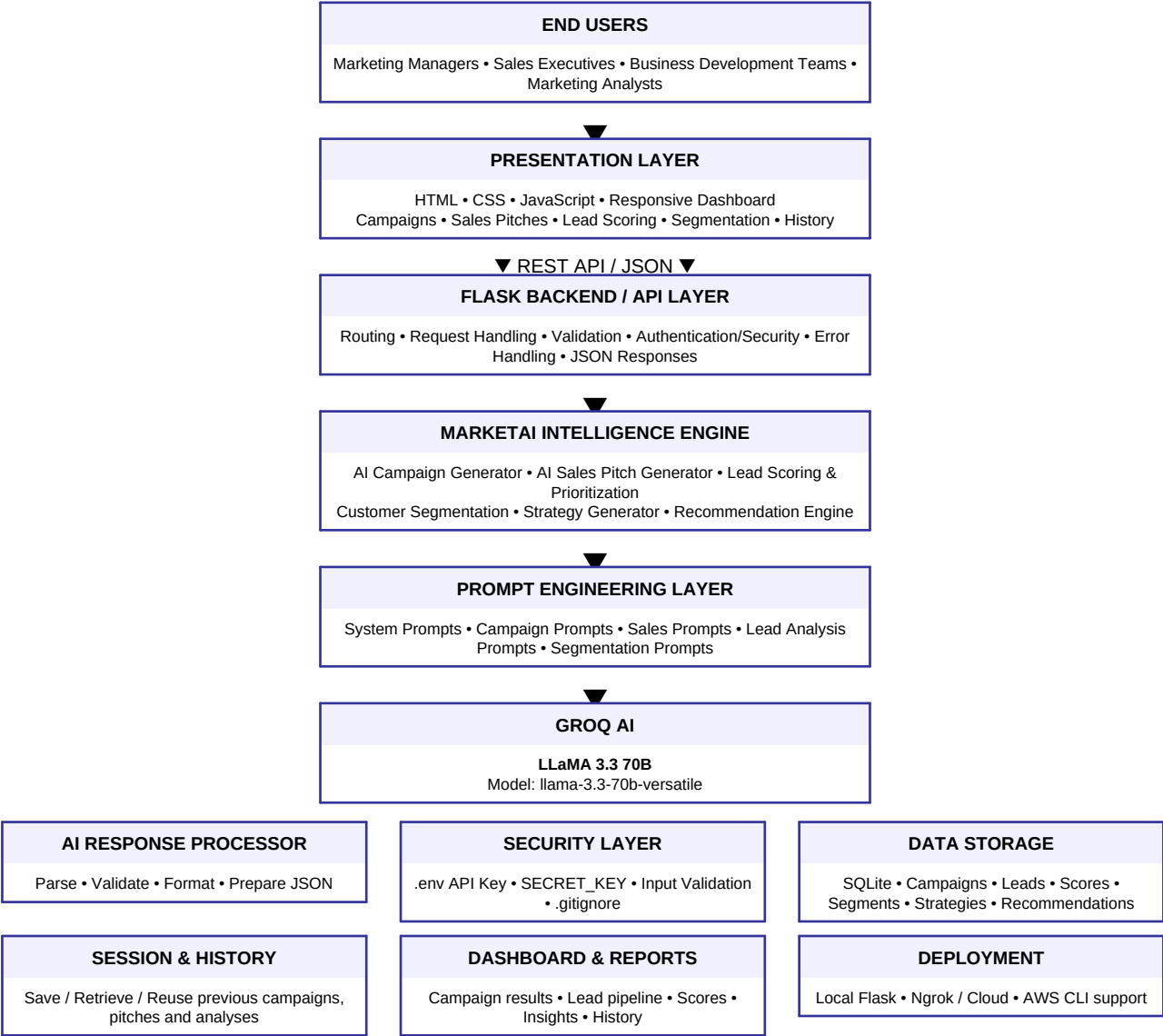


Solution Architecture – MarketAI Suite

Generative AI-Powered Sales & Marketing Intelligence Platform



1. Architecture Components

Layer	Technology / Component	Responsibility
Presentation	HTML, CSS, JavaScript	Responsive user interface and dynamic result rendering.
API	Flask REST API	Routes requests, validates input, calls services and returns JSON.
Business Logic	MarketAI Intelligence Engine	Coordinates campaign, sales, lead, segmentation and recommendation workflows.
AI	Groq LLaMA 3.3 70B	Generates context-aware marketing and sales content and analytical insights.
Prompt Layer	Prompt templates	Creates structured prompts using user and lead information.
Data	SQLite	Persists campaigns, pitches, leads, scores, strategies and history.
Security	.env / SECRET_KEY / validation	Protects credentials and validates application inputs.
Deployment	Ngrok / Cloud / AWS CLI	Provides public access and deployment support.

2. End-to-End Request Flow

1. User enters product, audience, objective, customer or lead information.
2. JavaScript sends the request to the appropriate Flask REST API endpoint.
3. Flask validates the request and invokes the relevant business service.
4. The service constructs a prompt using the Prompt Engineering layer.
5. Groq API sends the prompt to LLaMA 3.3 70B.
6. The AI response is parsed, validated and formatted.
7. The result is persisted in SQLite where appropriate.
8. Flask returns a JSON response to the frontend.
9. The dashboard displays the campaign, sales pitch, lead score or recommendation.
10. Previous results can later be retrieved from history without another AI generation request.

3. Scenario Mapping

Scenario	Architecture Flow	Output
Marketing Campaign	Frontend → Flask → Campaign Service → Prompt → Groq → Database	Slogans, social posts, emails, promotional content and strategy
Sales Pitch	Frontend → Flask → Sales Service → Prompt → Groq → Database	Pitch, value propositions, objection handling and talking points
Lead Scoring	CSV → Flask → Lead Service → Analysis → Scoring → Database	Lead score, High/Medium/Low priority and recommended action
History	Frontend → History API → SQLite → Dashboard	Previously generated campaign, pitch or lead analysis

4. Recommended Project Structure

```
MarketAI-Suite/  
  ├── app.py  
  ├── config.py  
  ├── requirements.txt  
  ├── .env  
  ├── routes/  
  │   ├── campaign_routes.py  
  │   ├── sales_routes.py  
  │   ├── lead_routes.py  
  │   └── history_routes.py  
  ├── services/  
  │   ├── ai_service.py  
  │   ├── campaign_service.py  
  │   ├── sales_service.py  
  │   └── lead_scoring_service.py  
  ├── prompts/  
  │   ├── campaign_prompt.py  
  │   ├── sales_prompt.py  
  │   ├── lead_prompt.py  
  │   └── segmentation_prompt.py  
  ├── models/  
  │   ├── campaign.py  
  │   ├── sales_pitch.py  
  │   ├── lead.py  
  │   └── history.py  
  ├── database/  
  │   ├── database.py  
  │   └── schema.py  
  ├── templates/  
  │   └── HTML pages  
  ├── static/  
  │   └── CSS and JavaScript  
  ├── tests/  
  │   └── automated tests  
  └── utils/  
      ├── validators.py  
      ├── error_handler.py  
      └── helpers.py
```

5. Security Architecture

The Groq API key and Flask SECRET_KEY are stored in environment variables and excluded from source control. All user inputs and uploaded CSV data should be validated. AI/API failures should be handled through centralized error handling. The backend, rather than the browser, communicates with the Groq API so that the API key is never exposed to the client.

6. Solution Architecture Summary

MarketAI Suite follows a layered architecture: the responsive web frontend communicates with Flask REST APIs; the Flask backend delegates requests to the MarketAI Intelligence Engine; prompt templates transform business inputs into AI instructions; Groq's LLaMA 3.3 70B generates the intelligence; results are validated, stored in SQLite and presented through the dashboard. The history layer enables reuse of previous outputs, while the security layer protects credentials and application data.